

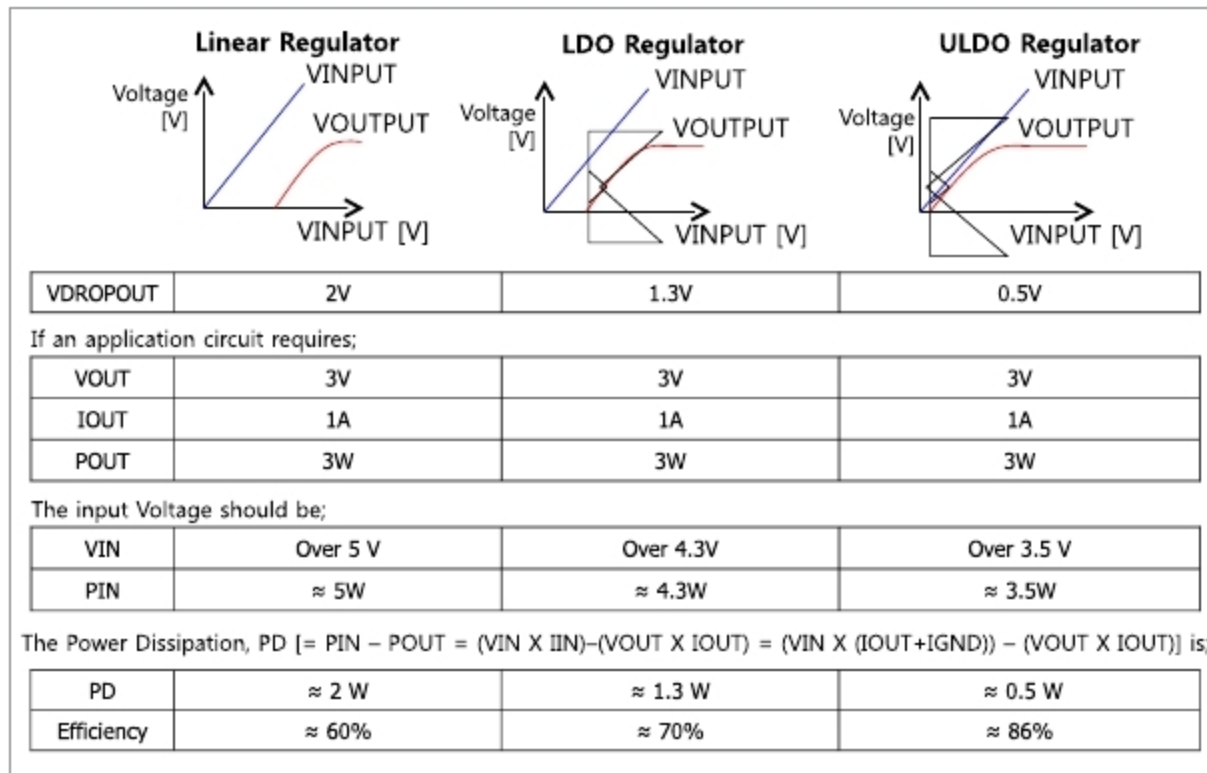


Fig. 2: Power conversion characteristics with respect to dropout voltage

Fig. 2 presents a comparison between power conversion characteristics with respect to dropout voltage performance of linear regulators. The presented three kinds of linear regulators need different minimum input voltage to get the same required output power. As shown in the figure, **lower dropout voltage causes better power conversion efficiency, resulting in lower power dissipation.**

Another advantage of lower power dissipation is lower heat occurrence that makes smaller package applications possible.

Pass elements

For ULDO regulators, PMOSFET or NMOSFET are used as the pass element. Their role is the same, but

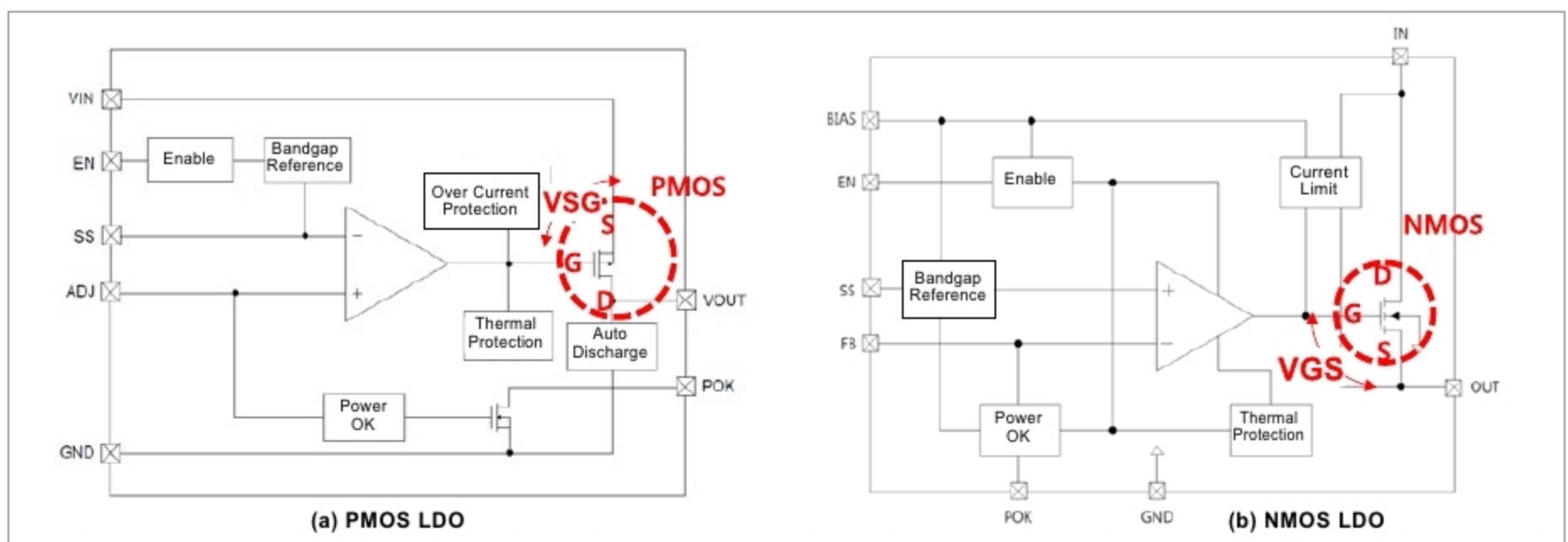


Fig. 3: Block diagrams of PMOS LDO and NMOS LDO

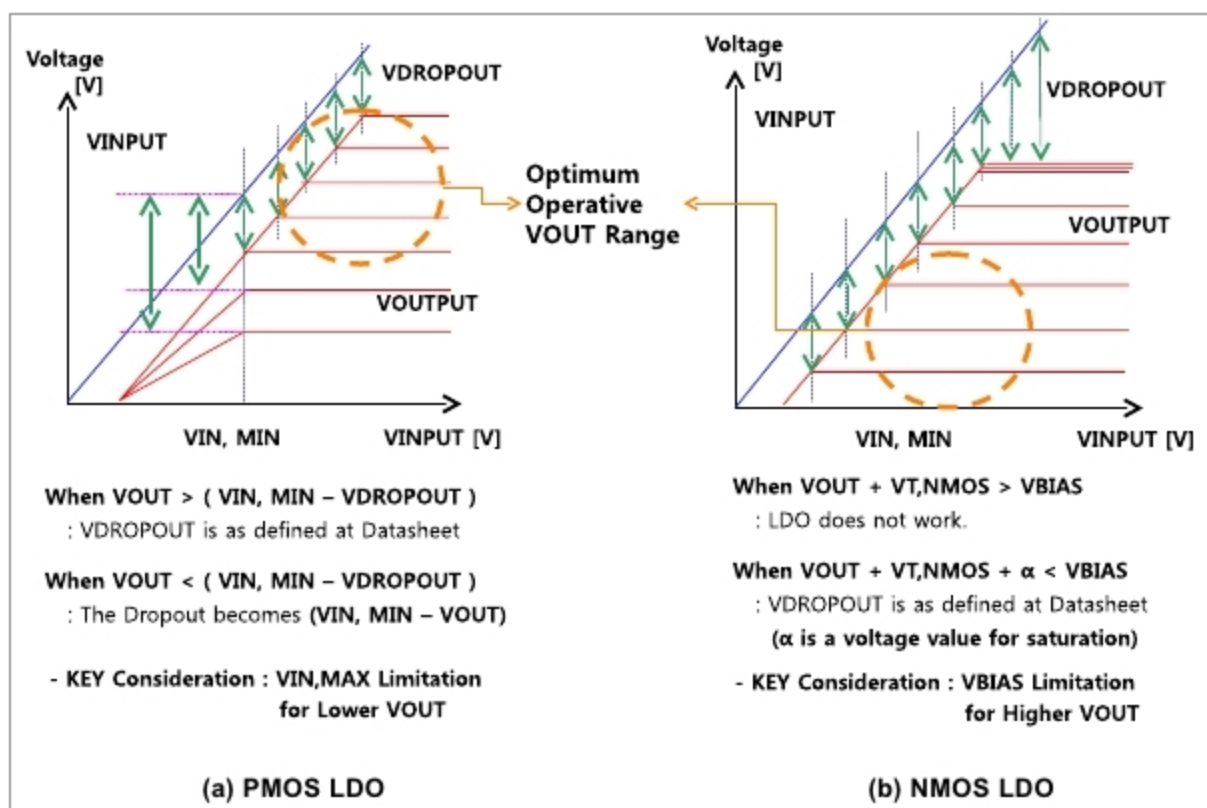


Fig. 4: Optimum operating range of ULDO

purpose of application is different. PMOSFET LDO has a typical LDO structure, as shown in Fig. 3(a). On-resistance performance (dropout voltage performance) is proportional to V_{SG} and it needs higher V_{IN} for better on-resistance performance. So, when output voltage is high, there is no problem in presenting good dropout performance.

However, as shown in Fig. 4(a), when output voltage gets low, dropout performance is restricted by minimum input voltage. So, PMOS LDO is suitable for relatively higher output voltage applications because dropout voltage becomes $[V_{IN, MIN} - V_{OUT}]$ when output voltage is smaller than $[V_{IN, MIN} - V_{DROPOUT, Typical}]$.

For example, for PMOS LDO with 0.5V of V_{DROPOUT} typical and 2.5V